

Example 3: Consider a mass on a spring, with relaxed length zero, in the x - y plane. The Lagrangian, $L = (m/2)(\dot{x}^2 + \dot{y}^2) - (k/2)(x^2 + y^2)$, is invariant under the change of coordinates, $x \rightarrow x + \epsilon y$ and $y \rightarrow y - \epsilon x$, to first order in ϵ (as you can check). Obtain the conserved current using Noether's theorem and interpret the result [Sol: $m(\dot{x}y - \dot{y}x)$].

Example. The spherical pendulum of the figure is allowed to rotate in three dimensions (with $0 \leq \theta < \pi$, $0 \leq \phi < 2\pi$).

- Find the Lagrangian of the system.
- Can you find a symmetry of this Lagrangian? If yes, use Noether's theorem to find the conserved quantity [Sol: $ml^2\dot{\phi}\sin^2(\theta)$].

